

Вспомогательное преобразование схемы определит комплексные параметры цепи

$$\omega = 2\pi f = 2\pi \cdot 50 = 314.1593 \text{ rad/s}$$

$$E_1 = \frac{E_{n1}}{\sqrt{2}} e^{j\varphi_1} = 17.678 e^{j25^\circ} \text{ V}$$

$$J_2 = \frac{J_{m2}}{\sqrt{2}} e^{j\varphi_2} = 10.607 e^{j15^\circ} \text{ A}$$

$$E_2 = \frac{J_2}{G_{m2}} = \frac{10.607 e^{j15^\circ}}{0.1} = 106.07 + j27.45 \text{ V}, R_{m2} = \frac{1}{G_{m2}} = \frac{1}{0.1} = 10 \Omega$$

$$E_{eq} = E_1 - E_2 = 17.678 e^{j25^\circ} - (106.07 + j27.45) = -88.39 - j19.98 \text{ V}$$

$$Z_{eq, in} = (R_{m1} + R_{m2}) + jX_{Lm1} = (5 + 10) + j2 = 15 + j2 \Omega = 15.13 e^{j7.59^\circ} \text{ deg}$$

Рассчитаем эквивалентные сопротивления ветвей приемника

$$X_{L1} = \omega L_1 = 314.1593 \cdot 0.006 = 1.885 \Omega, X_{C1} = \frac{1}{\omega C_1} = \frac{1}{314.1593 \cdot 0.000047} = 67.726 \Omega$$

$$X_{L2} = \omega L_2 = 314.1593 \cdot 0.11 = 34.558 \Omega, X_{C2} = \frac{1}{\omega C_2} = \frac{1}{314.1593 \cdot 0.000035} = 90.946 \Omega$$

$$X_{L3} = \omega L_3 = 314.1593 \cdot 0.07 = 21.991 \Omega, X_{C3} = \frac{1}{\omega C_3} = \frac{1}{314.1593 \cdot 0.000035} = 90.946 \Omega$$

$$Z_{eq} = 15.13 e^{j7.59^\circ} \cdot 4 \cdot 1.885 \cdot 67.726 = 41.1885 \cdot 67.726 = 2784.5 \Omega$$